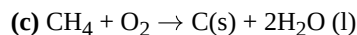


Solution

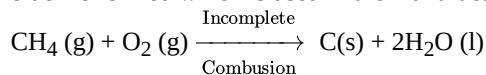
HYDROCARBONS WS

Class 11 - Chemistry

1.



Explanation: During incomplete combustion of alkanes (CH_4) with an insufficient amount of air or dioxygen (O_2), carbon black is formed which is used in the manufacture of ink, printer ink, black pigments and as filters. Thus,



2.

(a) lower than the boiling point of straight chain alkene.

Explanation: As the surface area decreases with branching, the boiling point also decreases.

3.

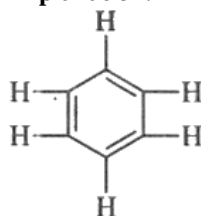
(d) Infinite

Explanation: Infinite

4.

(b) 12σ and 3π bonds

Explanation:



5.

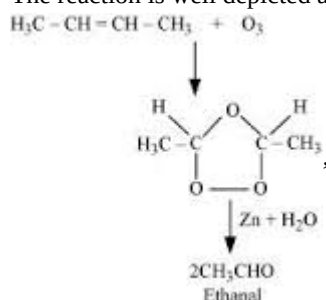
(b) CH_3CHO

Explanation: The given sequence of conversion steps represent Ozonolysis of 2-Butene, which follows the following path,

i. Formation of an unstable intermediate/ozonide (A)

ii. Cleavage of the intermediate/ozonide by ($\text{Zn} + \text{H}_2\text{O}$) to smaller molecules, giving out the compound B, which is **CH_3CHO (Ethanal)**

The reaction is well depicted as below,



6.

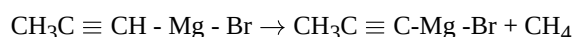
(b) Alkanes

Explanation: Alkyl halides react with dialkyl copper reagents to give alkanes. This is Corey-House reaction and gives good yields of both symmetrical and unsymmetrical alkanes.

7.

(d) CH_4

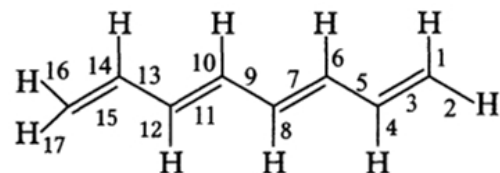
Explanation: Terminal alkynes act as a nucleophile. The organometallic compound 'Grignard's reagent (R-Mg-X), therefore, reacts to form alkane. Thus when $\text{CH}_3\text{Mg-Br}$ is treated with $\text{CH}_3\text{C}\equiv\text{CH}$ we get CH_4 as per the following equation:



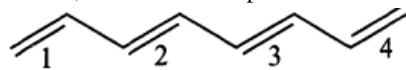
8. (c) $\text{HI} > \text{HBr} > \text{HCl}$
Explanation: $\text{HI} > \text{HBr} > \text{HCl}$
9. (c) Staggered
Explanation: In staggered conformation of ethane, any two H-atoms on adjacent C-atoms are as far apart as possible. As a result, the repulsions between the electron clouds of σ -bonds of two non bonded H-atoms are minimum and hence stability is maximum.
10. (c) Substitution reaction
Explanation: Substitution reaction (also known as single displacement **reaction** or single **substitution reaction**) is a chemical **reaction** during which one functional group in a chemical compound is replaced by another functional group.
11. (d) increases with increase in molecular mass.
Explanation: As molecular mass increases, the magnitude of Van der Waals forces of attraction increases and hence boiling point increases accordingly.
12. (b) one strong sigma (σ) bond due to head-on overlapping of sp^2 hybridised orbitals one weak pi (π) bond.
Explanation: Sigma bond is formed by the head-on overlap of sp^2 hybridized orbitals while the pi bond is formed by lateral or sideways overlapping of the two 2p-orbitals of the two carbon atoms. Alkenes contain 1 weak pi bond and 1 strong sigma bond.
13. (a) 1,3-dimethylbenzene
- Explanation:**

(1,3-dimethylbenzene)
14. (a) All of these
Explanation: A planar cyclic molecule having $(4n + 2)$ electrons is generally aromatic. Thus, all given options because of the presence of $6n$ electrons are aromatic.
15. (d) 3
Explanation: The number of chain isomers of C_5H_{12} are as follows:
- i. n-pentane $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$
 - ii. 2-Methylbutane $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_2\text{CH}_3$
 - iii. 2,2-Dimethylpropane $\text{CH}_3\text{C}(\text{CH}_3)_2\text{CH}_3$
16. (b) alkaline KMnO_4
Explanation: Baeyer's reagent is a dilute alkaline KMnO_4 solution. It is violet in colour when this solution reacts with the double bond the colour disappears and the whole solution becomes colourless.
17. (a) Delocalized π electron cloud
Explanation: Resonance stabilization explains the unusual stability of benzene. There is π electron delocalization in the ring which enhances the stability of the molecule.
18. (d) σ -Bond = 17
 π -Bond = 4

Explanation: A single bond is always a σ -bond and a double bond contains one σ and one π -bond.



Hence, 17 σ -bonds are present.



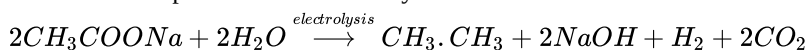
Hence, 4 π -bonds are present.

19.

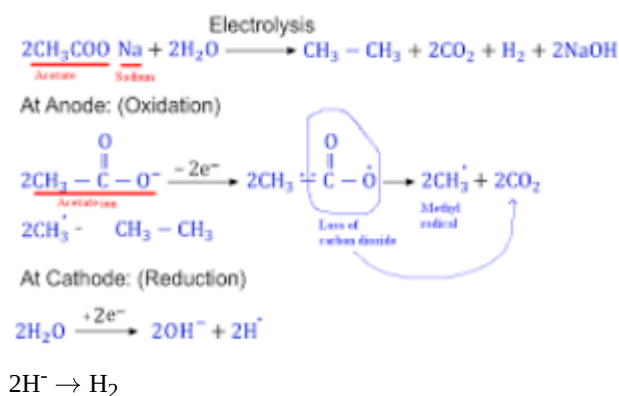
(b) Sodium acetate

Explanation:

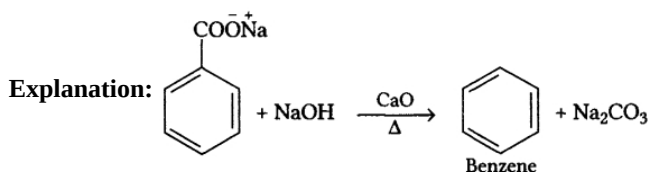
This is an example of Kolbe's electrolysis method. The reaction is:



The step-wise redox reactions occurring in the electrolytic cell are depicted as under



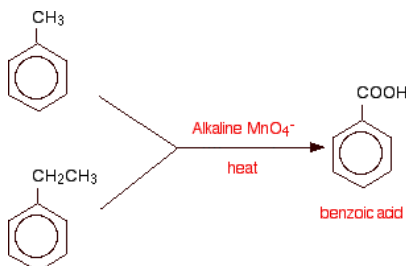
20. **(a) C₆H₆**



21. **(a) benzoic acid**

Explanation: All alkylbenzenes (such as C₆H₅CH₃ or C₆H₅C₂H₅) on oxidation with alkaline - KMnO₄

Therefore, the oxidation of ethylbenzene with alkaline KMnO₄ yields" Benzoic - acid.



22.

(c) 3,6 - Diethyl - 2 - methyloctane

Explanation: Following the rules of nomenclature, the IUPAC name of the given compound is 3, 6-Diethyl-2-methyloctane.

23.

(b) Carbon atoms form a an open chain.

Explanation: Aromatic hydrocarbon is not an open chain. They are ring structures. they contain one or more benzene rings either fused or isolated in their molecules.

24. (d) Substitution reaction
Explanation: Halogenation of alkanes is a substitution reaction because the hydrogen atoms are replaced by halogens in the presence of sunlight to give the corresponding alkyl halides. The reaction follows a free radical mechanism.
25. (b) Isolation from coal tar
Explanation: Since the isolation of benzene from coal tar involves the separation of benzene from other components through a physical operation "fractional distillation", it is not a synthetic method for the production of benzene.
26. (a) 1, 4-dibromobutane
Explanation: 1,4 - dibromobutane (ie. a dibromo derivative of an alkane, butane) when reacted within (Na – metal) ethereal solution produces alicyclic hydrocarbon, Cyclobutane as per the following reaction.
- $$\begin{array}{c}
 \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 \\
 | \qquad \qquad \qquad | \\
 \text{Br} \qquad \qquad \qquad \text{Br}
 \end{array}
 + 2 \text{Na} \xrightarrow{\text{ether}} \boxed{\phantom{\text{Cyclobutane}}} + 2 \text{NaBr}$$
- 1,4 dibromobutane Cyclobutane
- Therefore, the dibromo-derivative of an alkane is identified as 1, 4-dibromobutane.
27. (d) $\text{HO} - \text{CH}_2 - \text{CH} = \text{CH}_2$
Explanation: $\text{HO} - \text{CH}_2 - \text{CH} = \text{CH}_2$
28. (d) a mixture of benzene and $\text{Mg}(\text{OMe})\text{Br}$.
Explanation: This is because Grignard reagents react with proton from methanol to give benzene and $\text{Mg}(\text{OMe})\text{Br}$.
29. (b) CH_4
Explanation: Marsh gas which is also called methane is produced by the anaerobic bacterial decomposition of vegetable matter and the rumen of herbivorous animals underwater.
30. (d) $-\text{CI}^\ominus$
Explanation: $-\text{CI}^\ominus$ deactivate the benzene ring towards electrophilic substitution reaction.
31. (a) deactivates the ring towards electrophilic substitution.
Explanation: This is because the Nitro is an electron-withdrawing group, it pulls the electron density from the ring towards itself thereby decreasing the electron density in the ring and deactivating the ring towards attack by the electrophile.
32. (b) Benzyl chloride
Explanation: In the presence of light, the CH_3 group undergoes a substitution reaction to give benzyl chloride $\text{C}_6\text{H}_5\text{CH}_2\text{Cl}$ as the main product.
33. (a) chain isomers
- $$\begin{array}{c}
 \text{CH}_3 \\
 | \\
 \text{CH}_3 - \text{CH} - \text{CH}_3, \text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{CH}_3 \\
 \text{2 - methylpropane} \qquad \qquad \text{n - butane}
 \end{array}$$
- Molecular formula $\rightarrow$ Same
Chain of C-atom $\rightarrow$ Different
34. (c) $\text{CH}_3 - \text{C} \equiv \text{CH} + 2\text{HBr} \rightarrow$
Explanation: The reaction follows Markovnikoff's rule of addition which states that in addition of HBr to an unsymmetrical alkene, the major product is the one in which the Br gets attached to the more substituted carbon of the double bond.
35. (c) All of these
Explanation: Tobacco, coal and petroleum damages DNA of our body and causes cancer.

36.

(c) ∞

Explanation: ∞ can be rotated in any direction

37.

(a) benzene

Explanation: Phenol is reduced to benzene.

38.

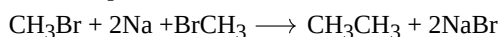
(c) 2-methylprop-1-ene

Explanation: $\overset{1}{\text{CH}_2} = \overset{2}{\underset{\text{CH}_3}{\text{C}}} - \overset{3}{\text{CH}_3}$
2-methylprop-1-ene

39.

(c) 2

Explanation: Wurtz reaction is used for the preparation of alkanes from alkyl halides in the presence of sodium and dry ether.
For example:

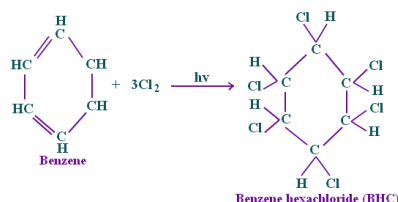


Thus two atoms of Na are used.

40.

(a) $\text{C}_6\text{H}_6\text{Cl}_6$

Explanation: Benzene, in the presence of sunlight and in absence of halogen carriers, undergoes an addition reaction with Cl_2 to form benzene hexachloride, $\text{C}_6\text{H}_6\text{Cl}_6$



41.

(a) $\text{R-Cl} < \text{R-Br} < \text{R-I}$

Explanation: As we move from I to Br to Cl, atomic size decreases, it becomes difficult to release the halogen as the bond length decreases. Hence I^- is the best leaving group among halogens and reacts the fastest.

42.

(b) 2-methylbut-3-yne

Explanation: $\text{H}_3\text{C} - \overset{3}{\underset{\text{CH}_3}{\text{CH}}} - \overset{2}{\text{C}} = \overset{1}{\text{CH}}$
3-methylbut-1-yne

43.

(d) Geometric isomers

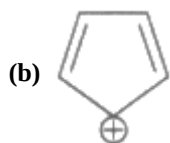
Explanation: Isomers which have the same structural formulae but differ in the relative spatial arrangements of atoms or groups around the double bond are called geometric isomers. Thus geometrical isomers are a type of stereoisomers.

44.

(b) C_2H_2

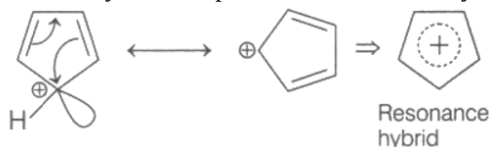
Explanation: Acetylene can be prepared by electrolysis of a concentrated solution of sodium or potassium salt of fumaric acid. This is Kolbe's electrolytic method.

45.

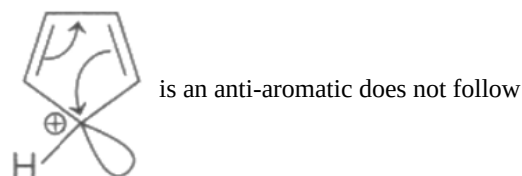


Explanation:

Aromaticity of a compound can be decided by Huckel's rule. In cyclopentadienyl cation (b), resonance takes place as follows:



Hence,



Huckel's rule as it has conjugated 4 π -electron ($4n\pi$, $n = 1$) system. Rest of the species are aromatic as each of them belongs to 6π -electron [$(4n + 2)\pi$, $n = 1$] system.

46.

(d) all of these

Explanation: all of these

47.

(c) Neopentane

Explanation: $(\text{CH}_3)_3\text{CCH}_3 + \text{HX} (\text{X}=\text{Cl}, \text{Br}, \text{I}) \rightarrow (\text{CH}_3)_3\text{CCH}_2\text{X}$

Hence a monosubstituted derivative is formed.

48.

(d) $\text{NaOH} + \text{CaO}$

Explanation: Soda lime = $\text{NaOH} + \text{CaO}$

49.

(d) A

Explanation: Because hydrocarbons contain both C and H and hence Hydrogen is not a hydrocarbon.

50.

(a) Pot. Succinate

Explanation: Ethene is obtained by the electrolysis of potassium succinate by the Kolbe's electrolytic method.

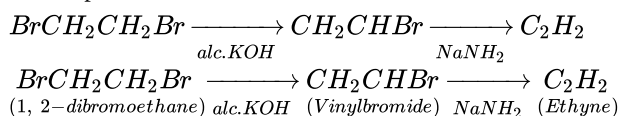
51.

(d) alcoholic KOH followed NaNH_2

Explanation: This is an example of elimination reaction in two steps.

In the first step alc. KOH is used as a base forming vinyl bromide ($\text{CH}_2 = \text{CHBr}$) and in the second step, a stronger base NaNH_2 (sodium -amide) is used.

The stepwise conversions reactions are,



52.

(a) Propene

Explanation: Elimination reaction takes place, a molecule of HBr gets eliminated to form propene.

53.

(c) A closed chain or a ring

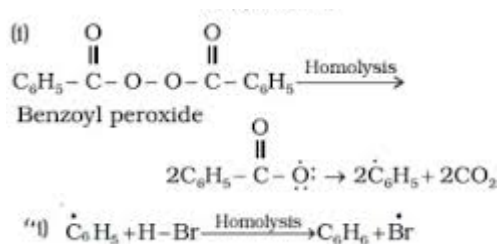
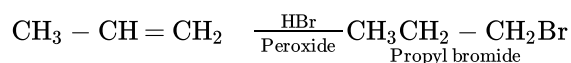
Explanation: Cycloalkanes contain single bond between the carbon atoms, they form ring structures and have the general formula C_nH_{2n} .

54.

(b) a free radical mechanism.

Explanation:

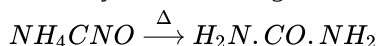
The mechanism involved along with the generation of a free radical [*i. e. phenyl free radical* ($\text{C}_6\text{H}_5\cdot$)] by the use of benzoyl peroxide, showing peroxide effect in given below,



55. (a) $3^\circ > 2^\circ > 1^\circ$

Explanation: This order of stability of carbocations is explained on the basis of dispersal of positive charge by the +I-effect and hyperconjugation effect of the alkyl groups. Greater the number of alkyl groups on the carbon atom carrying the positive charge, greater would be the dispersal of positive charge, and hence more stable will be the carbocation.

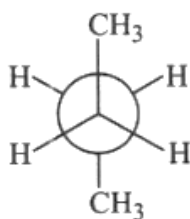
56. (c) Plants, animals, and synthetic methods
Explanation: Berzelius, a Swedish chemist proposed a theory termed as "Vital force theory", during the 16th century. According to the theory it was considered that organic compounds are formed by living bodies such as plants and animals. The theory thus differentiated between organic compounds and the inorganic compounds. Around the year 1780 attempts were made to synthesize or form organic compounds artificially by several chemists. However, this notion was rejected in 1828 when F. Wohler synthesized an organic compound, urea, from an inorganic compound ammonium cyanate.



Since then many organic compounds could be formed by synthetic methods also.

57. (d) Anti

Explanation:



Anti

58. (d) 3

Explanation: The three isomers will be ortho (i.e. 1,2- or 1,4-disubstituted), meta (i.e. 1,3- or 1,5-disubstituted) and para (1,4-disubstituted).

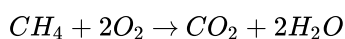
59. (c) both animal and vegetable kingdom

Explanation: Compounds which are isolated directly or indirectly from living organisms like plants and animals are called organic compounds.

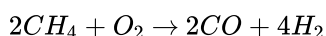
60. (d) trans-alkene
Explanation: trans-alkene

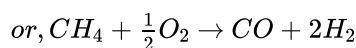
61. (c) $2\text{CH}_4(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{CO}(\text{g}) + 4\text{H}_2(\text{g})$

Explanation: CH_4 on complete combustion gets completely oxidized to form $(\text{CO}_2 + \text{H}_2\text{O})$, as per the following equation;



The given reaction,



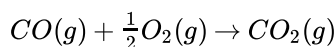


represents incomplete combustion, because stoichiometrically,

- for complete combustion 2 moles of O_2 per mole of CH_4 is required, while the only $\frac{1}{2}$ mole of O_2 per mole of CH_4 is taken to carry out its complete oxidation or combustion.
- $\text{O}_2(\text{g})$ is the limiting reagent for the reaction.

Thus, one mole of methane is burnt in a limited supply of oxygen, resulting in the formation of one mole of $\text{CO}(\text{g})$.

- $\text{CO}(\text{g})$ could have been further oxidized to form $\text{CO}_2(\text{g})$ as the final product of combustion.



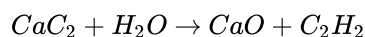
Hence, the given equation represents incomplete combustion of methane.

62.

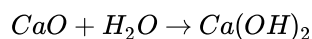
(b) CaC_2

Explanation: The reaction of CaC_2 (Calcium carbide) with water produces acetylene (C_2H_2).

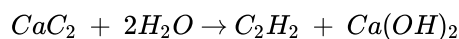
The chemical equation for the reaction of calcium carbide (CaC_2) with water to yield acetylene ($\text{HC} \equiv \text{CH}$) is



The calcium oxide (CaO) is formed as a byproduct, which simultaneously reacts with water to give $\text{Ca}(\text{OH})_2$



Thus, the overall reaction is,



63. (a) C_nH_{2n}

Explanation: Alkenes are unsaturated hydrocarbons with at least one double bond between the carbon atoms. Their general formula is C_nH_{2n} .

64.

(c) electrophilic addition reaction

Explanation: The electrophilic addition reaction between ethene and sulfuric acid. Alkenes react with concentrated sulfuric acid in the cold to produce alkyl hydrogensulphates. For example, ethene reacts to give ethyl hydrogensulphate.

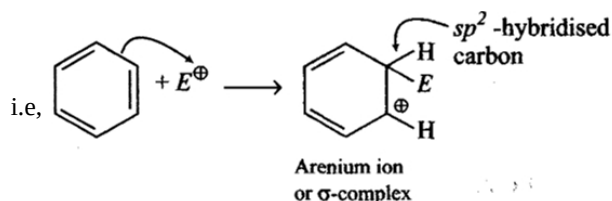
65. (a) A mixture of low boiling hydrocarbons.

Explanation: Pyrolysis is the decomposition of a compound by the application of heat. It involves the breaking of C-C and C-H bonds resulting in the formation of lower boiling alkanes.

66.

(c) Both σ Complex and arenium ion

Explanation: Attack of electrophile on benzene nucleus results in the formation of σ -complex or arenium ion, in which one of the carbon is sp^2 -hybridised.

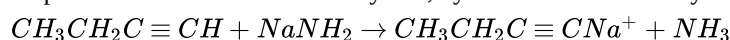


67.

(c) $\text{BrCH}_2\text{CH}_2\text{CH}_2\text{CH}_3$ and $\text{CH}_3\text{CH}_2\text{C} \equiv \text{CH}$

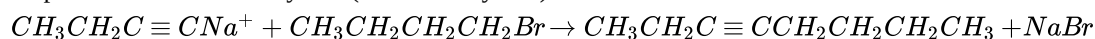
Explanation: 3-Octyne can be synthesized as per the following stepwise conversion reactions:

Step 1. Formation of a Sodium alkynide, by the reaction of 1-butyne with sodium amide



1-butyne sodium amide sodium butynide

Step 2. Reaction of the alkynide (sodium butynide) with 1-bromobutane



Sodium Butynide 1-bromobutane 3-Octyne

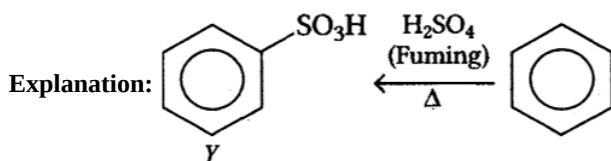
68.

(c) C_nH_{2n}

Explanation: Cycloalkane with one benzene ring has a formula C_6H_{12} .

$\therefore$ The formula becomes C_nH_{2n}

69. (a) $C_6H_5SO_3H$



70.

(b) methane

Explanation: By Wurtz reaction, higher alkanes are prepared.

i.e. $R - Cl + 2Na + R - Cl \rightarrow R - R + 2NaCl$

Thus, methane cannot be prepared by this reaction.

71. (a) Electrophilic addition

Explanation: Alkynes have two pi bonds, hence they are electron rich species. They attract electrophiles thereby undergoing electrophilic addition reactions.

72.

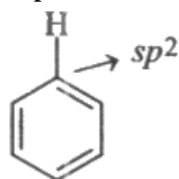
(d) reduction process

Explanation: reduction process as hydrogen atom has been added.

73.

(c) sp^2 - hybridised

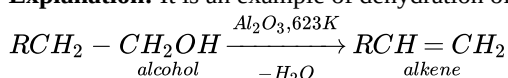
Explanation:



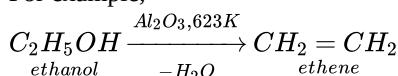
74.

(c) C_2H_4

Explanation: It is an example of dehydration of alcohols to give alkenes.



For example,



Thus, when ethanol vapours are passed over heated alumina, the alcohol gets dehydrated to form ethene, as given above.

75.

(c) cis-but-2-ene

Explanation: Cis form of alkene is found to be more polar than the trans form e.g. dipole moment of cis-but-2-ene is 0.33 Debye, whereas, dipole moment of the trans form is almost zero or it can be said that trans-but-2-ene is non-polar.

